Cisco Packet Tracer Project

Step-by-Step Implementation Guide

Course: Computer Networks | CC431
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Scenario: University Campus Network

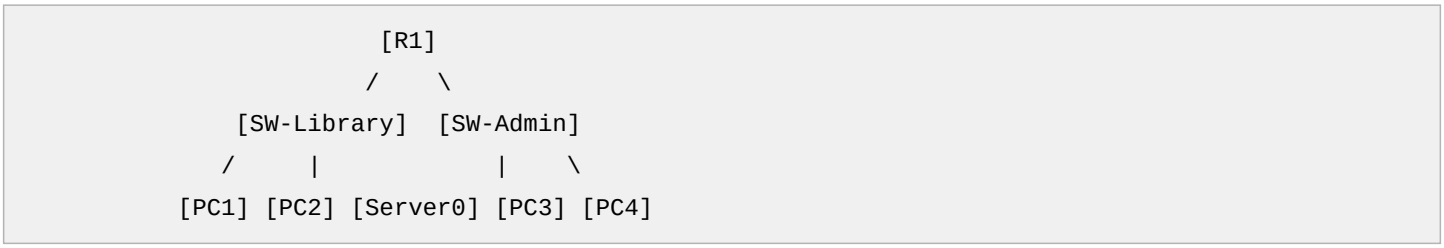
STEP 1 -- Scenario & Network Topology

Scenario: A University Campus with two buildings -- Library and Admin Block -- connected through a central Router.

Devices Required in Packet Tracer:

Device	Name	Quantity
Router	R1	1
Switch	SW-Library	1
Switch	SW-Admin	1
PC	PC1, PC2	2 (Library)
PC	PC3, PC4	2 (Admin)
Server	Server0	1

Topology Diagram:



STEP 2 -- IP Addressing Table

Two subnets are used to separate the Library and Admin networks:

Device	Interface	IP Address	Subnet Mask	Default GW
R1	Fa0/0	192.168.1.1	255.255.255.0	---
R1	Fa0/1	192.168.2.1	255.255.255.0	---
SW-Library	VLAN1	192.168.1.2	255.255.255.0	192.168.1.1
SW-Admin	VLAN1	192.168.2.2	255.255.255.0	192.168.2.1
Server0	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	Fa0	192.168.1.20	255.255.255.0	192.168.1.1
PC2	Fa0	192.168.1.21	255.255.255.0	192.168.1.1
PC3	Fa0	192.168.2.20	255.255.255.0	192.168.2.1
PC4	Fa0	192.168.2.21	255.255.255.0	192.168.2.1

Subnet 1 (Library): 192.168.1.0 / 24

Subnet 2 (Admin): 192.168.2.0 / 24

STEP 3 -- Open Packet Tracer & Add Devices

- From the Bottom Panel choose Routers --> drag "1841" or "Router-PT"
- Choose Switches --> drag "2960" twice
- Choose End Devices --> drag 4 PCs and 1 Server

Connect Cables:

Connection	Cable Type
PC / Server <--> Switch	Copper Straight-Through
Switch <--> Router	Copper Straight-Through
Router <--> Router	Serial DCE (if using 2 routers)

STEP 4 -- Configure PCs & Server IPs

For each PC: Click PC --> Desktop tab --> IP Configuration --> enter IP, Subnet Mask, Default Gateway from the table above.

For Server0: IP = 192.168.1.10 | Mask = 255.255.255.0 | GW = 192.168.1.1

STEP 5 -- Router Configuration (CLI Commands)

TIP: Click on Router --> CLI tab --> press Enter --> type the commands below.

```
Router>  enable
Router#  configure terminal

! --- Hostname ---
Router(config)#  hostname R1

! --- Enable Secret Password ---
R1(config)#  enable secret cisco123

! --- Console Password ---
R1(config)#  line console 0
R1(config-line)#  password console123
R1(config-line)#  login
R1(config-line)#  exit

! --- Banner Message ---
R1(config)#  banner motd # Welcome to University Network - Authorized Access Only #

! --- Interface Fa0/0  (Library subnet) ---
R1(config)#  interface FastEthernet0/0
R1(config-if)#  ip address 192.168.1.1 255.255.255.0
R1(config-if)#  no shutdown
R1(config-if)#  exit

! --- Interface Fa0/1  (Admin subnet) ---
R1(config)#  interface FastEthernet0/1
R1(config-if)#  ip address 192.168.2.1 255.255.255.0
R1(config-if)#  no shutdown
R1(config-if)#  exit

! --- Static Route (only needed if using 2+ routers) ---
! R1(config)#  ip route 192.168.3.0 255.255.255.0 <next-hop-IP>

! --- Save Configuration ---
R1(config)#  exit
R1#  write memory
```

STEP 6 -- Switch Configuration (CLI Commands)

SW-Library:

```
Switch>  enable
Switch#  configure terminal
Switch(config)#  hostname SW-Library
```

```
SW-Library(config)#  enable secret switch123
SW-Library(config)#  line console 0
SW-Library(config-line)#  password console123
SW-Library(config-line)#  login
SW-Library(config-line)#  exit
SW-Library(config)#  banner motd # Library Switch - Authorized Only #
SW-Library(config)#  interface vlan 1
SW-Library(config-if)#  ip address 192.168.1.2 255.255.255.0
SW-Library(config-if)#  no shutdown
SW-Library(config-if)#  exit
SW-Library(config)#  ip default-gateway 192.168.1.1
SW-Library(config)#  exit
SW-Library#  write memory
```

SW-Admin: (same steps, different IP)

```
Switch>  enable
Switch#  configure terminal
Switch(config)#  hostname SW-Admin
SW-Admin(config)#  enable secret switch123
SW-Admin(config)#  line console 0
SW-Admin(config-line)#  password console123
SW-Admin(config-line)#  login
SW-Admin(config-line)#  exit
SW-Admin(config)#  banner motd # Admin Switch - Authorized Only #
SW-Admin(config)#  interface vlan 1
SW-Admin(config-if)#  ip address 192.168.2.2 255.255.255.0
SW-Admin(config-if)#  no shutdown
SW-Admin(config-if)#  exit
SW-Admin(config)#  ip default-gateway 192.168.2.1
SW-Admin(config)#  exit
SW-Admin#  write memory
```

STEP 7 -- Server Services Configuration (3 Services)

Click on Server0 --> go to the Services tab --> configure each service below.

7.1 DNS Service

- Select DNS from the left panel
- Turn DNS Service ON
- Add new record: Name = www.university.com | Type = A | Address = 192.168.1.10
- Click Add

7.2 HTTP Service

- Select HTTP from the left panel
- Make sure HTTP is turned ON
- Optionally edit index.html (e.g., type "Welcome to University Campus")

7.3 Email Service (SMTP / POP3)

- Select EMAIL from the left panel
- Turn SMTP and POP3 ON
- Set Domain Name = university.com
- Add users:
- Username: user1 Password: pass1 --> click Add
- Username: user2 Password: pass2 --> click Add

STEP 8 -- Connectivity Testing

8.1 Ping Test

On PC3: Desktop --> Command Prompt --> type:

```
ping 192.168.1.20
```

Expected result: Reply from 192.168.1.20 ... (4 replies = success)

8.2 Traceroute Test

On PC3: Desktop --> Command Prompt --> type:

```
tracert 192.168.1.10
```

You will see the path: PC3 --> R1 (192.168.2.1) --> Server0 (proves routing works)

8.3 DNS + HTTP Test (Web Browser)

On any PC: Desktop --> Web Browser --> type in URL bar:

```
www.university.com
```

Expected: The university server webpage opens successfully.

8.4 Email Test

- On PC1: Desktop --> Email --> Configure:
- Your Name: User1 | Email: user1@university.com
- Incoming / Outgoing Server: 192.168.1.10

- Username: user1 | Password: pass1
- Click Save, then Compose and send an email to user2@university.com
- Check that user2 receives the email.

STEP 9 -- Documentation Checklist

Take a screenshot of each item below and include them in your project report:

```
[ ]  Full network topology screenshot
[ ]  IP Addressing Table  (use the table from Step 2)
[ ]  Router CLI configuration  (scroll up to capture all commands)
[ ]  Switch CLI configuration for both switches
[ ]  DNS service settings on Server0
[ ]  HTTP service settings on Server0
[ ]  Email service settings on Server0
[ ]  Successful Ping result  (4 replies)
[ ]  Successful Traceroute result
[ ]  Web browser opening www.university.com
[ ]  Email send and receive confirmation
```

BONUS -- Optional (+2 Marks)

Option A: DHCP on Router (instead of Server)

```
R1(config)#  ip dhcp pool LIBRARY-POOL
R1(dhcp-config)#  network 192.168.1.0 255.255.255.0
R1(dhcp-config)#  default-router 192.168.1.1
R1(dhcp-config)#  dns-server 192.168.1.10
R1(dhcp-config)#  exit

! Exclude static IPs from DHCP pool
R1(config)#  ip dhcp excluded-address 192.168.1.1 192.168.1.20

! Then on each PC: IP Configuration  -->  select DHCP  (instead of Static)
```

Option B: ACL (Access Control List)

```
! Example: Block PC4 from accessing the Server
R1(config)#  access-list 10 deny host 192.168.2.21
R1(config)#  access-list 10 permit any
R1(config)#  interface FastEthernet0/0
R1(config-if)#  ip access-group 10 out
```

Option C: Connect Multiple Networks (Second Router)

- Add Router R2 connected to R1 via a Serial link
- Assign a third subnet (e.g., 192.168.3.0/24) to R2
- Add Static Routes on both R1 and R2 pointing to each other's networks

Option D: Professional Topology & Documentation

- Label all devices with clear, descriptive names
- Use a consistent IP address scheme
- Write a clean project report with all screenshots organized
- Include a short paragraph describing your chosen scenario

Troubleshooting Tips

Problem	Likely Cause	Solution
Ping fails	Interface is down	Run: no shutdown on the interface
Ping fails	Wrong default gateway	Recheck gateway on PC / Switch
Wrong cable light	Wrong cable type	Use Straight-Through for PC-Switch
DNS not resolving	DNS server IP wrong	Set DNS server = 192.168.1.10 on PC
Webpage not loading	HTTP service is off	Enable HTTP on Server services tab
Email not sent	Wrong SMTP server IP	Use Server IP, not the domain name

IMPORTANT -- Correct Order of Steps:

1. **Add all devices and connect cables first**
2. **Set IP addresses on all PCs and Server**

3. Configure the Router (interfaces + security + save)
4. Configure both Switches (hostname + security + save)
5. Enable Server services (DNS, HTTP, Email)
6. Run all tests and capture screenshots
7. Write your documentation / project report